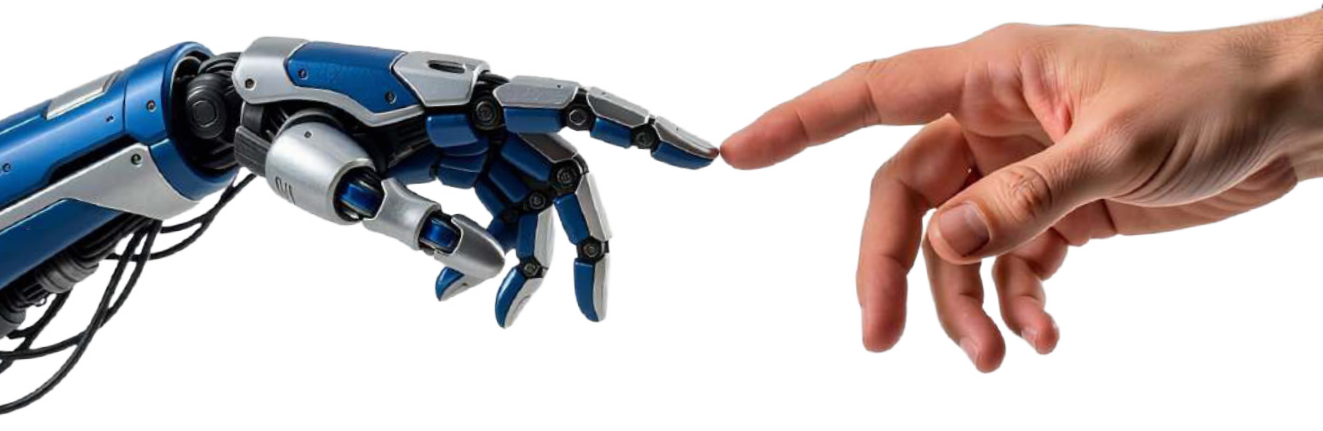
VIDYASHILP UNIVERSITY

Bachelor's Program in

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

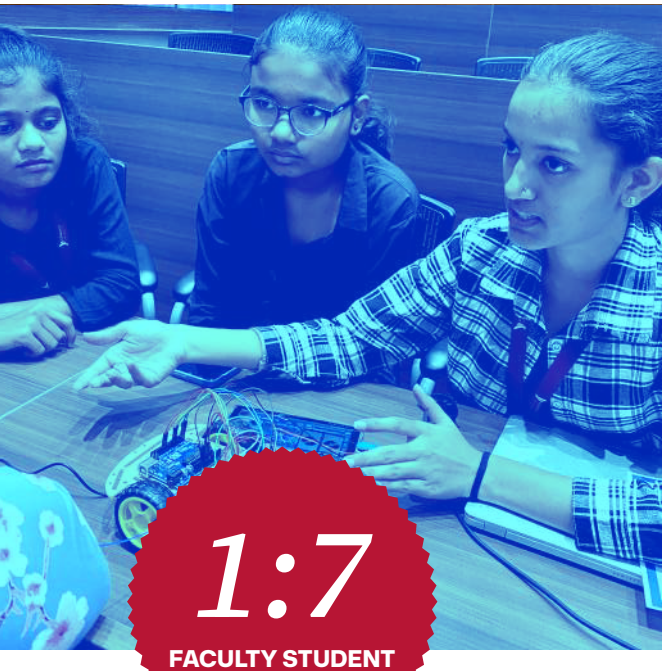
B.Tech. (Hons.) in Computer Science & Engineering - AI/ML



VIDYASHILP
SCHOOL OF ENGINEERING & TECHNOLOGY

About the Program

Shaping Real-World AI/ML Wizards at the Intersection of Data & Technology to Drive Impactful Industry Solutions



B.Tech. (Hons.)
in Computer Science & Engineering - AI/ML

AI Innovators for the Real World

- Advanced Machine Learning and Neural Networks
- AI Model Optimization and Deployment
- Cloud-based AI Systems

Intelligent Systems and Technology

- Deep Learning and Generative AI
- Natural Language Processing and Computer Vision
- Edge AI and Embedded Intelligence

Future-Ready Industry Applications

- Autonomous Systems and Robotics
- AI-driven Product Development
- Ethical AI and Responsible Innovation

Lead Global Innovation with World-Class Faculty

Our distinguished faculty have advanced academic training from some of the world's top Universities, including the IISc - Bengaluru, IIT- Bombay, BITS - Pilani, IIT- Dhanbad , IIM-Kozhikode and Bharti Vidyapeeth University, Pune, among others. Learn from their expertise, and embark on your own journey of educational transformation.

Why Choose the B.Tech. (Hons.) in Computer Science & Engineering (AI/ML) at Vidyashilp University?

Our B.Tech. (Hons.) in Computer Science & Engineering (AI/ML) Program is thoughtfully crafted to equip students with both the theoretical foundations as well as the practical skills needed to become successful professionals in this exciting and most impactful technology of the 21st century.

Cutting-Edge Curriculum:
Integrates Computer Science and Engineering with advanced topics like Artificial Intelligence, Machine Learning, Deep Learning, Reinforcement Learning, and Natural Language Processing.

Customizable Learning Pathways:
Students have the freedom to personalize their academic journey through a range of Elective Courses for a focused approach. Interdisciplinary minors in areas like Psychology, Design, Marketing, Finance, Law, and Economics further expand career possibilities, enabling students to operate at the intersection of data and domain knowledge. The open academic system fosters unique combinations of skills tailored to individual ambitions.

Electives:
A key highlight of the program is the flexibility to choose from a diverse range of elective courses, such as Generative AI models, Parallel Computing, Probabilistic Graphical Models, Bayesian Methods in Machine Learning, Multi-Modal Learning, Data Visualization and Story Telling, Full Stack Development, Backend Systems and API Development, Big Data Analytics, Web Frameworks, Mobile Application Development, Performance Optimization & Testing, and Advanced Statistical Analysis.

Interdisciplinary Opportunities:
Broaden career prospects with opportunities to pursue minors in diverse fields such as Psychology, Design, Marketing, Finance, Law, and Economics.

Holistic Skill Development:
Emphasis on soft skills such as communication, critical thinking, and teamwork.

Hands-On Learning:
Real-world projects, internships, and practical sessions ensure industry readiness.

Expert Faculty:
Learn from accomplished academicians and industry professionals with expertise in computational sciences and cutting-edge technologies.

State-of-the-Art Infrastructure:
Access to modern labs and tools for practical and research work.

Global Perspective:
Curriculum designed to address international challenges and equip students for global careers.

Future-Ready Graduates:
Prepares students for roles in AI, data-driven decision-making, and innovative technology solutions.



Course Distribution Across the Program

The undergraduate journey for students enrolled in the B.Tech. Computer Science & Engineering (AI/ML) program at Vidyashilp University is meticulously crafted to seamlessly integrate the University Core, Program Foundation, and Major discipline courses, fostering a transformative and enriching educational experience.

University Core

Equips students with Communication Skills, Learning to Learn, Environmental Studies and Sustainability, Fundamentals of Indian Constitution & Human Rights, Community Engagement, Socio-Cultural Perspectives on Indian Life, Critical Thinking, and Problem Solving for Design Thinking.

Program Foundation

Establishes a strong base in Language, Smart devices & Smart Materials, Introduction to Quantum Mechanics, Design Workshop, Multivariate Calculus, Probability and Statistics, Linear Algebra, and Programming in Python.

Program Core

Focuses on practical and rigorous learning across various disciplines of computer science including Data Structures, Theoretical Computer Science, Analysis of Algorithms, Digital Design and Computer Organization, Object Oriented Programming, Microprocessors and Computer Architecture, Operating Systems, Networks, Databases Management, Optimization Techniques, Machine Learning, and Artificial Intelligence.

Program Honors

Offers scholarly rigor in key areas such as Deep Learning, Natural Language Processing, Reinforcement Learning, Image Processing & Computer Vision, Cloud Computing and AI Model Deployment.

Electives

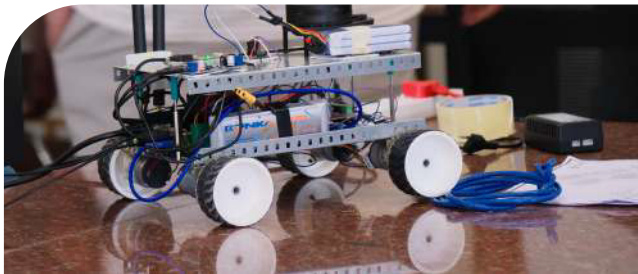
Generative AI models, Parallel Computing, Probabilistic Graphical Models, Bayesian Methods in Machine Learning, Multi-Modal Learning, Data Visualization, Full Stack Development, Backend Systems, Big Data Analytics, etc.

Minor Discipline

Students can pursue Minors in diverse domains, fostering interdisciplinary knowledge and innovation.

Research and Internship

Students also undertake Internships with leading companies gaining valuable industry exposure and practical experience.



A Comprehensive Overview

University Core

Critical Thinking | Problem Solving for Design Thinking | Communication Skills | Learning to Learn | Environmental Studies and Sustainability | Fundamentals of Indian Constitution and Human Rights | Community Engagement | Socio-Cultural Perspectives on Indian Life.

Program Foundation

Language, Smart devices & Smart Materials, Introduction to Quantum Mechanics, Design Workshop, Multivariate Calculus, Probability and Statistics, Linear Algebra, and Programming in Python.

Program Core

Data Structures, Theoretical Computer Science, Analysis of Algorithms, Digital Design and Computer Organization, Object Oriented Programming, Microprocessors and Computer Architecture, Operating Systems, Networks, Databases Management, Optimization Techniques, Machine Learning, and Artificial Intelligence.

Program Honors

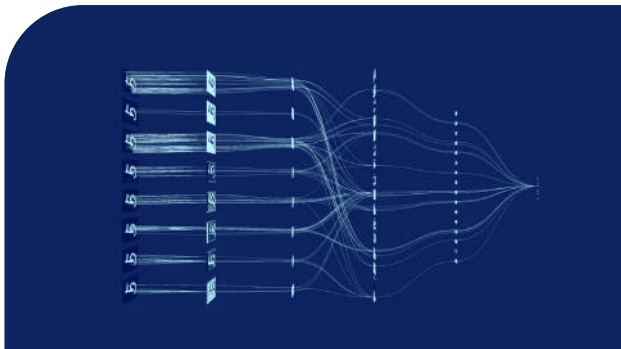
Deep Learning, Natural Language Processing, Reinforcement Learning, Image Processing & Computer Vision, Cloud Computing and AI Model Deployment.

Minors

Economics | Psychology | Design | Marketing | Finance | Law

Electives

Generative AI models, Parallel Computing, Probabilistic Graphical Models, Bayesian Methods in Machine Learning, Multi-Modal Learning, Data Visualization and Storytelling, Full Stack Development, Backend Systems and API Development, Big Data Analytics, Web Frameworks, Mobile Application Development, Performance Optimization & Testing, and Advanced Statistical Analysis & Research.



Faculty



P S Sastry
Dean, School of Engineering & Technology
PhD, IISc, Bengaluru



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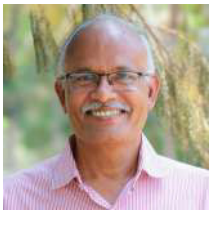
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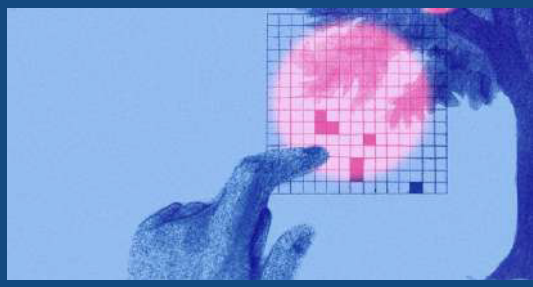
Saroj Snehal Shivagunde
Assistant Professor
School of Engineering
& Technology
PhD (CSE), IIT,
Guwahati

KNOW MORE ABOUT OUR FACULTY

Learning Journey Across Academic Years

Year 1: Foundation

Acquire a robust foundation in Mathematics and Computer Science by learning topics such as Calculus, Linear Algebra, Data Structures and Programming, analytical problem solving skills etc.



Year 2: Exploration

Gain a good background in Computer Science by learning topics such as Computer Organization, Algorithms and advanced programming concepts. Get a good understanding of the field by learning basics of Artificial Intelligence, Machine Learning, statistical methods, and Data Analysis techniques.

Year 3: Specialization

Dive deeper into AI/ ML by studying topics such as Deep Learning, Computer Vision, Natural Language Processing, ML System Development, Cloud Computing etc.

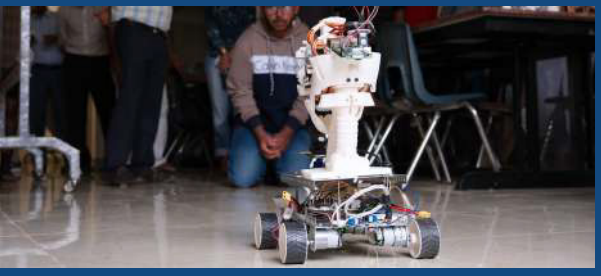


Year 4: Proficiency

Acquire technical competencies through advanced elective courses, capstone project and industry internships.

Achievements







AUTOWN'24 at BITS Pilani

Under the expert guidance of faculty, Dr. D.C.Kiran and Dr. N. Mehala, two teams from B.Tech 2nd year program, earned 1st and 3rd prizes respectively, for their outstanding technological solutions.

Bengaluru Mobility Challenge

A collaborative team of students and faculty secured first place at IISc in the Bengaluru Mobility Challenge for developing AI-based objection, utilizing AI-based object detection models and VU's advanced GPU machines. By processing over 9 lakh images and identifying vehicles across 2000+ camera feeds, the team demonstrated cutting-edge innovation and outstanding teamwork.

IEEE Leadership Summit 2024

Competing against 80 teams globally, our B.Tech. team clinched first place with Meera, a patient support robot known for its innovative design and functionality, reflecting the technical expertise and creativity nurtured at Vidyashilp University.

Basha Techathon by Bhashini and Google Cloud

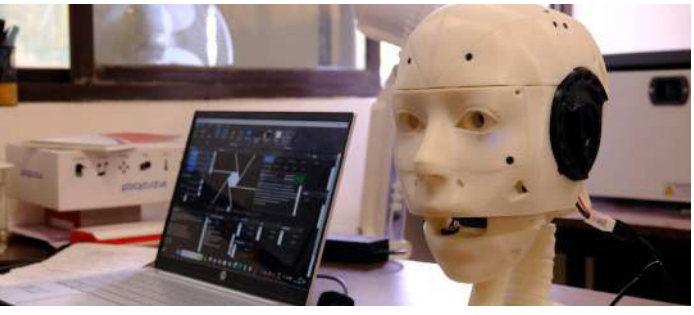
Our Data Science team received recognition for their AI-driven solutions to optimize the Indian Railways' complaint management system:

* Multi-modal AI Innovation was recognized as a Top Three Finisher.
* Advanced Voice Analysis AI was awarded with Special Jury's Recognition.



Shilpites Showcase

AI/ML students at Vidyashilp University work on challenging projects that help them develop strong analytical skills and a deeper understanding of the domain. The projects focus on solving real-world problems through data-driven decisions.



VURA

VURA combines the power of speech and image recognition, showcasing the innovative spirit of students at VU. By integrating AI, machine learning, and natural language processing, it highlights how students apply cutting-edge technologies to create interactive, real-world solutions.

FORENSIC SKETCHING ROBOT

The AI-powered forensic sketching robot combines facial landmark detection, speech-to-text, and cloud computing to address the limitations of traditional forensic sketching. By providing an interactive, AI-enhanced system, it allows for more precise suspect identification, reducing human error and bias while ensuring greater forensic accuracy and credibility.

FORENSIC VISION

Leveraging computer vision and UV light, Forensic Vision enhances crime scene evidence detection by reducing human error and bias. It offers a secure, transparent process for evidence logging and filing, ensuring tamper-proof records and the legal admissibility of forensic data while improving the accuracy of investigations.



ULTRASONIC DISTANCE SENSOR

Redefining precision in measurement, the Ultrasonic Distance Sensor enhances applications in Autonomous Navigation, Healthcare Robotics, and Marine Research, ensuring accurate data for critical operations.

AUTOMATED FIREFIGHTER

Designed for rapid response and pinpoint accuracy, the Automated Firefighter is set to revolutionize firefighting and search and rescue missions, improving search and rescue in high risk environments.

Internship Highlights

Here are a few of our student internships



*Scholarship * - up to 100%*

Profile-Based Scholarships to Encourage Diversity

*Subject to University Scholarship Policy

- P. Dayananda Pai Scholarship
- SAT scholarships
- Talent-based scholarships
- Need-based scholarships
- Dean's Honors Scholarship
- Other Categories

28+ crore awarded in scholarships since 2022



Admissions

1. *Submission of the application*
2. *Evaluation of the application based on the academic and non-academic profiles*
3. *Interaction with academic counselor and expert committee*
4. *Portfolio and document evaluation if eligible for a scholarship*
5. *Result notification with the LOS*
6. *Admission confirmation after the payment of acceptance fee and fulfilment of eligibility criteria*

Know More

I was particular about finding a university that values hands-on experience over textbooks. That's why I chose VU. Here they blend theoretical knowledge with real world applications, giving us a head start in our careers. In fact, we've already designed and built our own robot which is one of the many projects on campus.

Muhammad Kashif Haidery,
B.Tech. (Hons.)



Message from the Dean

Our B.Tech (Hons.) in Computer Science & Engineering (AI/ML) Program is strategically aligned with the demands of the global market. It is especially designed to train the next generation of innovators who will harness the transformative power of Artificial Intelligence and Machine Learning. At Vidyashilp University, you won't just learn algorithms; you will learn how to apply them to solve real-world problems.

P S Sastry
Professor & Dean

Apply Now

VIDYASHILP UNIVERSITY

VIDYASHILP SCHOOL OF ENGINEERING & TECHNOLOGY

Vidyashilp University Founding Campus
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Vidyashilp University is established under Karnataka Act No. 20 of 2021